

Solar Site Suitability Assessment

LOCATION: Kern County, CA — Solar Star / Edwards-Sanborn corridor · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.1 mi, 220 kV)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	91	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	93	NREL PVWatts v8	HIGH
Terrain slope	0.17	95	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH
Land cover type	0.08	60	National Land Cover Database 2021	HIGH

Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	50	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Methodology

FRAMEWORK

GIS-based Multi-Criteria Decision Analysis (GIS-MCDA)

- Doorga et al. (2019) — Renewable and Sustainable Energy Reviews 104:133-146
- Hernandez et al. (2015) — PNAS 112:13579-13584
- Al-Shammari et al. (2026) — Renewable Energy (in press)
- Charabi & Gastli (2011) — Renewable Energy 36(9):2554-2561
- Ong et al. (2013) — NREL/TP-6A20-56290

WEIGHT CALIBRATION

Constrained optimization (scipy SLSQP) within literature-supported bounds against EIA Form 860 ground truth, per NERC region.

DATA SOURCES

14 criteria (9 scored, 5 exclusion) from federal and open data sources, with per-criterion source selection via the data-selection engine and quality-ordered data trees.

KNOWN LIMITATIONS

- NWI wetlands data unavailable nationally (SSURGO hydric-soil proxy used).
- EJScreen data is a static 2024 snapshot (EPA tool discontinued).
- Scoring does not assess interconnection capacity or queue position.

DISCLAIMER: This assessment is based on publicly available federal data and peer-reviewed methodology. It is intended for screening purposes and does not replace site-specific field investigation, interconnection studies, or environmental surveys.